

Exploratory Data Analysis (EDA) Report

Task 2: Exploratory Data Analysis & Business Intelligence

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Objective

The objective of this task is to analyze the sales dataset using SQL and Exploratory Data Analysis (EDA) techniques to identify trends, patterns, and business insights. A static dashboard was also created to present the key performance indicators (KPIs).

Dataset Overview

- Dataset Name: ApexPlanet_DataAnalytics_Dataset
- Total Records: 980
- Database: sales_db
- Table Name: apexplanet_dataanalytics_dataset

The dataset contains sales transaction details such as Order ID, Customer ID, Customer Name, Age, Gender, City, Product, Category, Quantity, Price, and Total Sales.

SQL Analysis Performed

The following SQL operations were performed:

- Displayed sample records
 - Counted total orders
 - Calculated total revenue
 - Calculated average order value
 - Calculated average customer age
 - Identified top 5 products by revenue
 - Calculated sales by category
 - Calculated sales by city
 - Calculated sales by gender
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Key Findings

- Total Orders: 980
 - Total Revenue: ₹135,910,843.18
 - Average Order Value: ₹138,684.53
 - Average Customer Age: 41.36 years
 - Highest Selling Product: Mobile
 - Highest Revenue Category: Electronics
 - Highest Revenue City: Patna
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Data Visualization

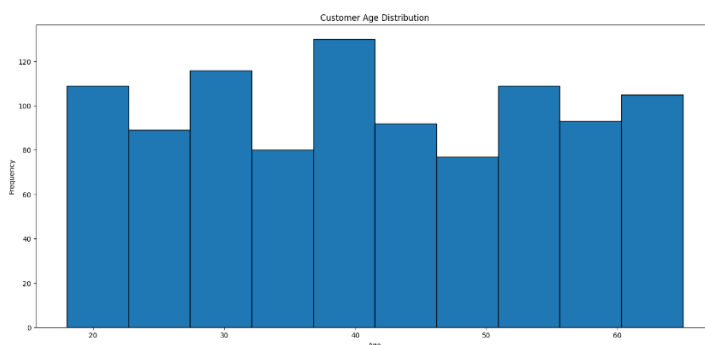
The following visualizations were created:

- Histogram
- Bar Chart
- Scatter Plot
- Correlation Heatmap
- Pair Plot
- Excel Dashboard

These visualizations helped identify relationships between variables and understand sales performance.

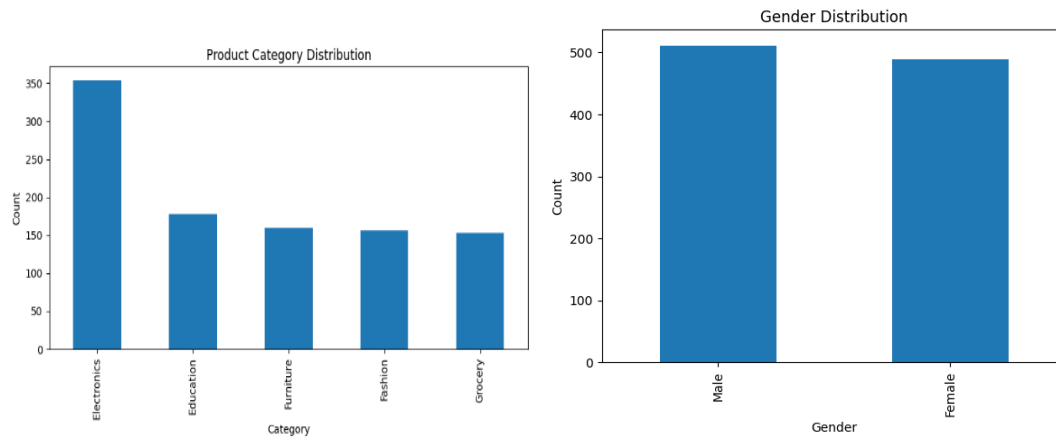
Customer Age Distribution

The customer age ranges from 18 to 65 years, with most customers falling between 30 and 50 years.



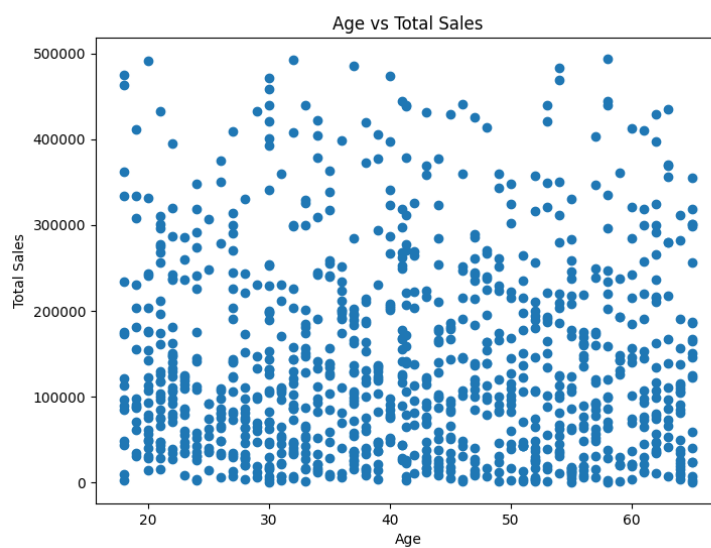
Product Category Distribution

Electronics is the most purchased category, followed by Education.



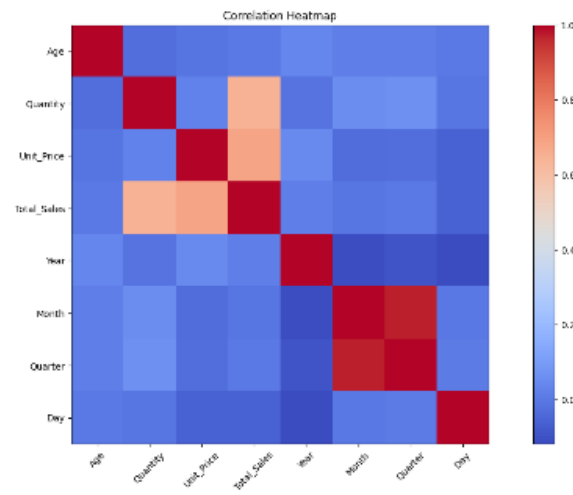
Scatter Plot

There is no strong relationship between customer age and total sales.



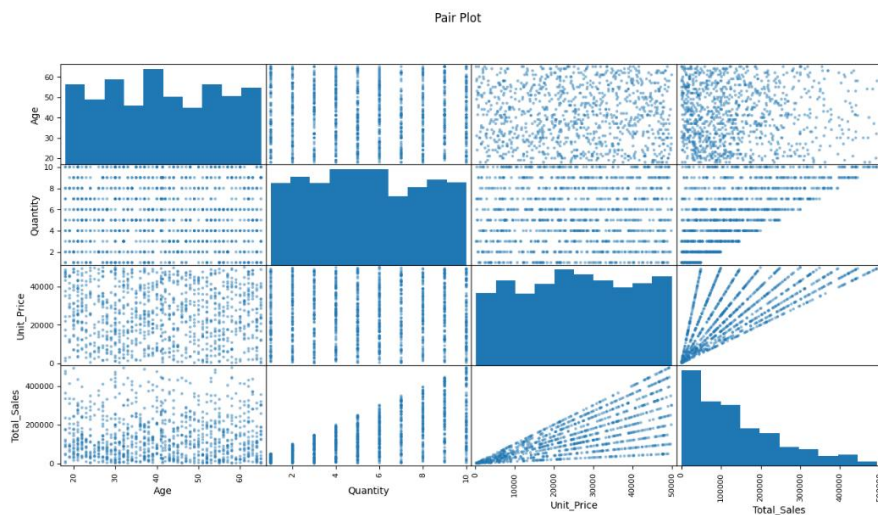
Heatmap

Unit Price and Total Sales show a strong positive correlation.

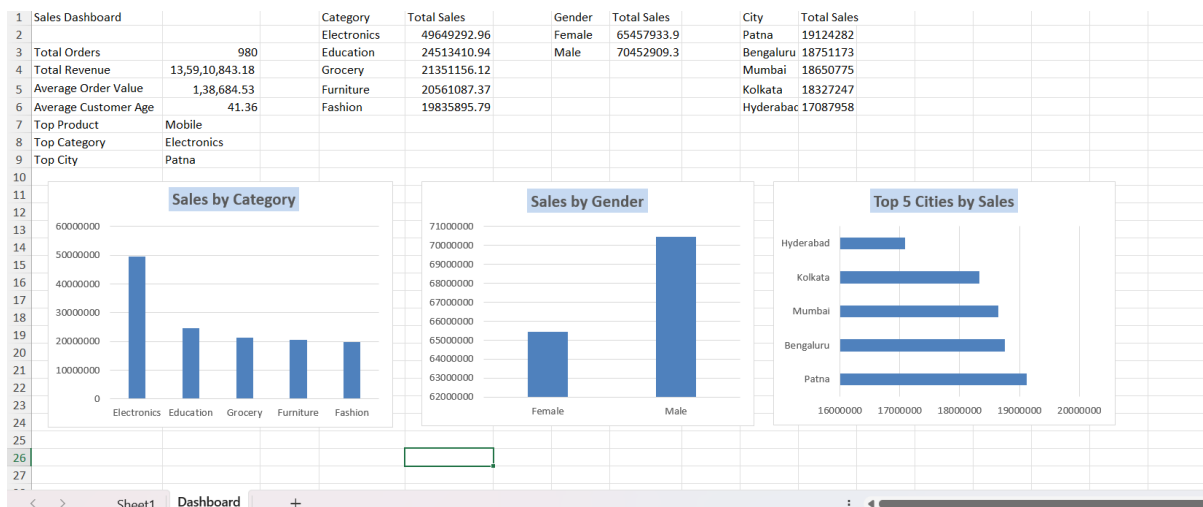


Pair Plot

The pair plot compares numerical variables simultaneously and confirms the relationships among Quantity, Unit Price, and Total Sales.



Excel Dashboard



Dashboard KPIs

The dashboard includes:

- Total Orders
- Total Revenue
- Average Order Value
- Average Customer Age
- Top Product
- Top Category
- Top City
- Sales by Category Chart
- Sales by Gender Chart
- Top 5 Cities by Sales Chart

Conclusion

The Exploratory Data Analysis successfully identified important business insights from the sales dataset. SQL queries helped answer business questions, while visualizations made trends easier to understand. The dashboard provides a clear summary of key business metrics and can support decision-making.